Heart Disease Risk Analysis

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Abstract

This report describes the results of a data investigation into heart disease using real patient data from the University of California Irvine (UCI) Heart Disease Dataset. The investigation aimed to identify the key factors associated with heart disease and develop a computer program that can predict the likelihood of a patient having heart disease. The investigation was conducted in three stages: cleaning and preparing the data, exploring the patterns in the data using charts, and developing a machine learning program to make predictions. The final program was able to correctly identify heart disease in 83 out of every 100 patients it had never seen before. It was also very effective in correctly identifying patients who do have heart disease. In fact, the program was able to correctly identify 89 out of every 102 patients who have heart disease. The key area for improvement is the 13 patients who have heart disease but were not identified by the program.

Methodology

The dataset used in this analysis is the UCI Heart Disease Dataset, one of the most widely used benchmarks in medical machine learning research. It is a multivariate dataset — meaning it captures many different measurements per patient simultaneously — making it well-suited to uncovering complex relationships that no single factor could reveal on its own. The data comes from four hospitals across the United States, Hungary, and Switzerland and contains clinical records for over 900 patients. Each record includes 13 measurements taken during a routine examination, such as age, cholesterol, blood pressure, and the type of chest pain reported.

Each patient record contains the following measurements:

Attribute	Full Name	Plain-Language Description
age	Age	Patient's age in years
sex	Sex	Biological sex (Male / Female)
cp	Chest Pain Type	Category of chest pain reported (0–3)
trestbps	Resting Blood Pressure	Blood pressure at rest (mm Hg)
chol	Serum Cholesterol	Cholesterol level in blood (mg/dL)
fbs	Fasting Blood Sugar	Whether blood sugar > 120 mg/dL after fasting
restecg	Resting ECG Results	Heart electrical activity at rest (0–2)
thalch	Max Heart Rate Achieved	Highest heart rate recorded during exercise test
exang	Exercise-Induced Angina	Whether chest pain occurred during exercise
oldpeak	ST Depression (Oldpeak)	Drop in heart signal during exercise vs. rest
slope	Slope of Peak ST Segment	Shape of the heart signal during peak exercise
ca	Number of Major Vessels	Major blood vessels visible on fluoroscopy (0–3)
thal	Thalassemia	Type of blood disorder detected
target	Heart Disease (Outcome)	Lael: 0 = No disease, 1 = Disease present

Data Cleaning

The following actions were taken to make the data reliable:

Missing data — There were some data points that had missing information in the data set. In this case, the median of the data point was used.

Simplification of the data — In the original data set, the data on the level of heart disease had values ranging from 0-4. This data was then simplified to a yes/no condition, with 0 indicating the absence of the disease and 1 indicating the presence of the disease.

Conversion of text data to numerical data — The data on sex and the type of pain were in text format and had to be converted to numerical data.

Removal of irrelevant data — The data on the patient and the hospital were removed because it did not have any medical implications on the aim of the study.

Analysis

The analysis was performed in Jupyter Notebook with major libraries such as pandas and NumPy for data manipulation and analysis, Matplotlib and Seaborn for data visualization, and scikit-learn for building and testing the model.

Data Preprocessing: The data was preprocessed to ensure that the target variable was transformed into a binary variable (presence or absence of heart diseases), irrelevant columns were removed, and missing values were replaced using the median function. Encoding of categorical variables was performed to ensure that all variables could be used for machine learning.

Exploratory Data Analysis (EDA) was performed to understand patterns within the data. For instance, a violin plot for age distribution based on gender and heart diseases, a stacked bar chart for types of chest pain and heart diseases, and a joint plot for cholesterol and maximum heart rate.

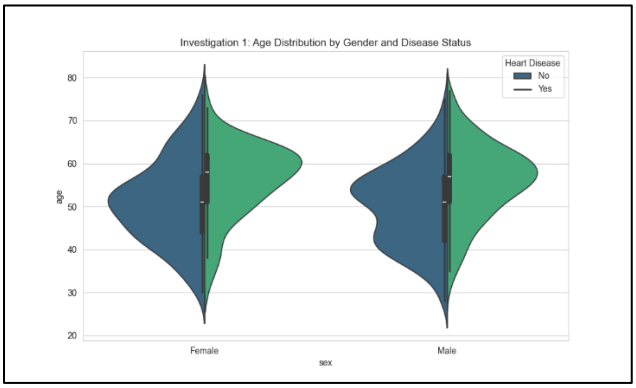
For prediction purposes, the data was divided into training and testing sets (80% and 20% respectively) through stratified splitting. Then, a Random Forest Classifier with 150 trees was used to create a model based on the data. Ensemble learning was used to improve the accuracy of the model.

Lastly, feature importance was used to identify variables that contributed significantly to predicting heart diseases.

Effect of Age and Gender

The violin chart below shows the distribution of the patients by their age, considering the gender and the presence or absence of heart disease.

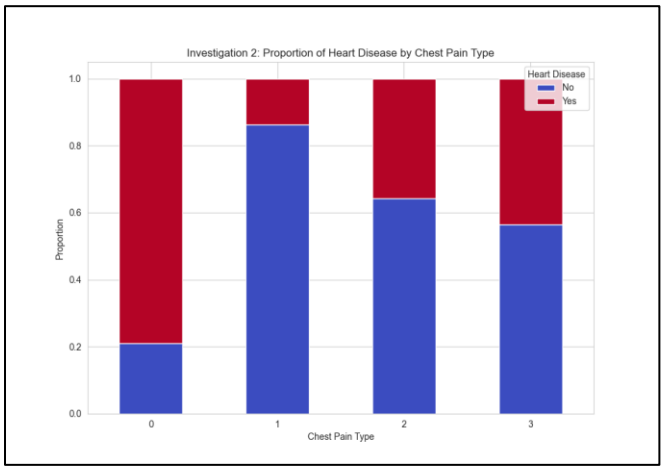
The greater the width of the section in the shape, the larger the number of patients at the particular age. It is evident that heart disease is not limited to the elderly. Even though the cases of the disease among women are concentrated between the ages 60 and 65, among men the cases of the disease cut across the entire age spectrum, including the younger population in their 40s and 50s.



Effect of chest pain

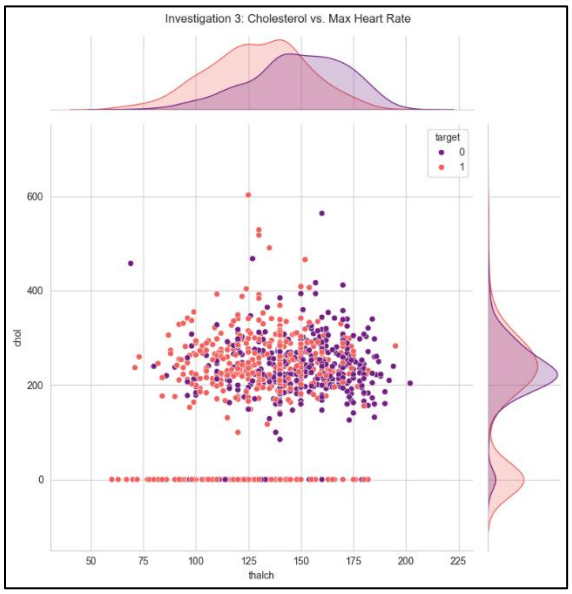
This bar chart indicates the percentage of patients who went on to have heart disease confirmed for each of the different types of chest pain. Blue = No disease; Red = Disease. The results are surprising. The patients experiencing typical angina (Type 0 – the 'classic' heart pain) have the largest percentage of those experiencing heart disease – around 80%. But what is even more surprising is that those patients who did not experience any chest pain (asymptomatic – Type 3) also have a percentage of those experiencing heart disease of around 45%.

Almost half of all patients experiencing heart disease did not have any chest pain. Therefore, relying on chest pain as a screening tool will result in a significant percentage of patients being overlooked.



Effect of maximum heartrate and cholesterol level

In this graph, each data point represents a patient's cholesterol level versus his maximum heart rate attained during a physical workout test. Points marked purple represent patients without any form of disease, whereas those marked red/orange represent patients who have some disease.

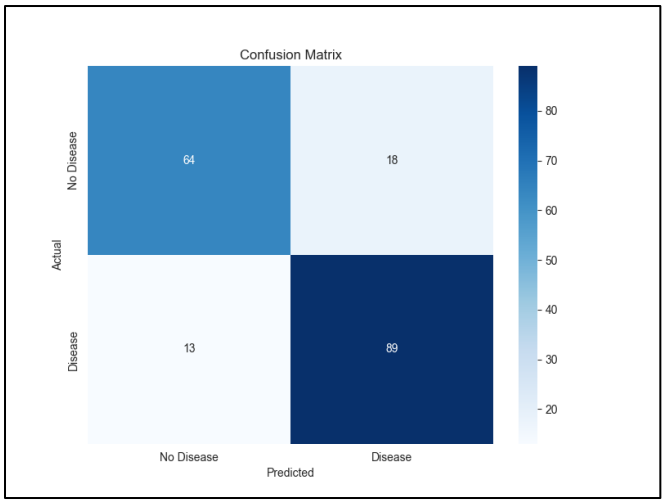


The most obvious trend evident from this graph is heart rate and not cholesterol levels. Diseased patients seem to have a low maximum heart rate during a physical workout as evidenced by the fact that the average heart rate for diseased patients peaks at 130-140 heartbeats per minute as opposed to 155-160 for non-diseased patients. This observation makes a lot of physiological sense since diseased hearts find it hard to pump strongly when stressed.

Confusion Matrix

The outcomes of the Machine learning model is presented as a confusion matrix. It shows that the model correctly predicted the 64 healthy patients as 'No Disease' and the 89 diseased patients as 'Disease'. However, the model flagged 18 healthy patients as false positives and 13 diseased patients as false negatives. Here the false negatives are the most serious as these patients would go undiagnosed and could suffer serious health consequences.

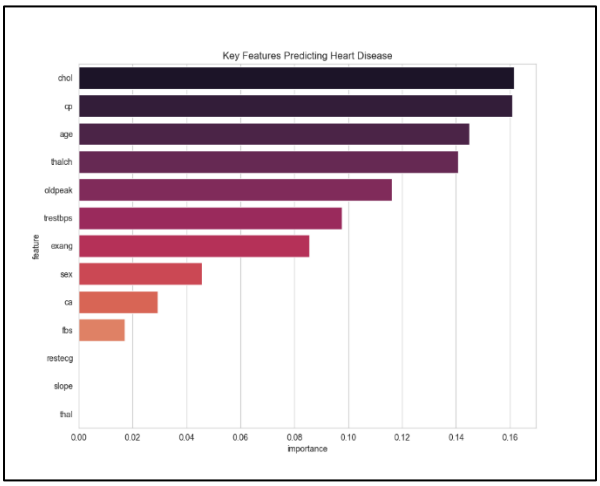
The model achieved an overall 83% accuracy. In a real life clinical setting, priority would be to reduce the 13 missed cases.



Most important factors

After training, the model indicated which of the variables were of particular significance for predicting:

- Cholesterol and type of chest pain — the two most significant predictor variables, each providing about 16% of the prediction model's basis.
- Age and maximum heart rate — not far behind at around 14–15%, thus validating the trends observed above.
- ST depression (oldpeak) and blood pressure — exercise-induced stress variables that provide valuable information.
- REST ECG, Slope, and Thalassemia — provided very little input into the prediction model, likely as a result of missing data.
- A single variable does not predict heart disease. The strength of the prediction algorithm lies in the use of 13 different variables combined, just like a good physician considers the entire range of symptoms.



Conclusion

It is clear that machine learning can be used in order to help find people who might have heart disease by providing good baseline results along with the finding that cholesterol, chest pain type, age, and exercise heart rate are the most valuable information in this dataset.

The research can be improved further by :

- Finding ways to reduce false negatives - This may include trying such methods as altering the decision threshold or cost-sensitive learning.
- Testing on other data - The model needs to be validated through its testing on data from other patients who were not included in the dataset provided.
- Potential use as a clinical tool - At least 83% accuracy makes the algorithm worth using as a preliminary filter for risky cases.

The report is written for general readership. As such, any technical terms used were purposely simplified. For more detailed explanation of methods and codes used, see the attached Jupyter notebook named HAnalysis.ipynb.